MATLAB Tutorial

MATLAB (matrix laboratory) is a numerical computing environment and fourth-generation programming language. Developed by MathWorks, MATLAB allows matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, and interfacing with programs written in other languages, including C, C++, Java, and Fortran. MATLAB will be used as an interface to send and receive data to/from the FPGA as well as to understand some image processing concepts before implementation in the FPGA.

1. Download `matlab-fix.sh.txt` file and change its name to `matlab-fix.sh` . Open a terminal window and execute it. This needs to be done only one time and will allow us to work with serial port in MATLAB.
2. Plug in the FPGA board to the PC by connecting the USB cable to connector J6 on the board.
3. Open a terminal and navigate to the project folder. Load the MATLAB module and run MATLAB.

   ```
   $ module load matlab/2022a
   $ matlab&
   ```

4. Create a new script by clicking the *New Script* button as shown in Figure 1.

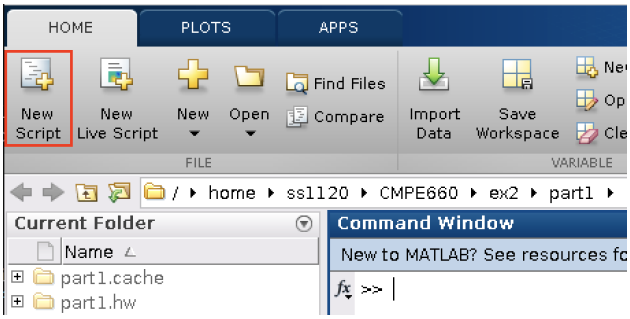


Figure 1: New script button.

4. A new *Editor* window will open as shown in Figure 2.

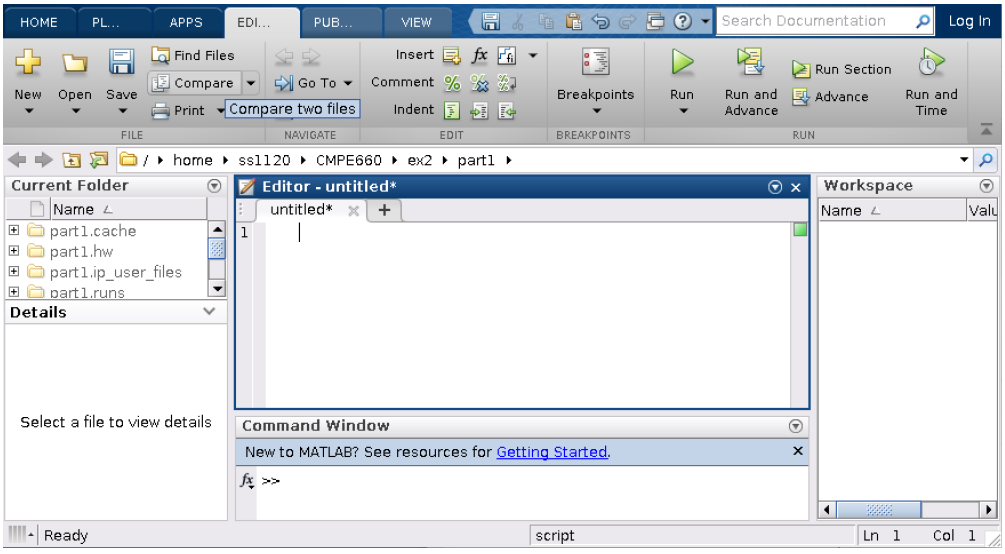


Figure 2: The editor window.

5. (a) The following code shows how to transmit the characters 'abc' one after another with a second pause in between each. There is also a commented out line for reading characters back from the FPGA which can be used for Part 2. Enter the code into the MATLAB Editor window (the code is also given on MyCourses).

```
%clear existing variables, close any figure windows, clear the command line
close all
clear all
clc

% Clear any existing serial ports
delete(instrfind);

% Create a serial port object
serialPort = serial('/dev/ttyUSB1', 'BaudRate', 115200, 'DataBits', 8, 'Parity', 'none',
'StopBit', 1, 'InputBufferSize', 3, 'OutputBufferSize', 3);
% TODO Modify InputBufferSize and OutputBufferSize to match, or be larger than the data.
% They control how much data Matlab will buffer before sending to (or receiving from)
% the UART

% open the serial port for reading/writing
fopen(serialPort);

% send characters to the FPGA
fprintf(serialPort,'%c','a');
pause(1)
fprintf(serialPort,'%c','b');
pause(1)
fprintf(serialPort,'%c','c');

% read 3 characters from the FPGA
% fscanf(serialPort,'%c%c%c',3)

%close the serial port
fclose(serialPort);
```

   (b) Instead of sending predefined characters, it is often useful to be able to send data from a file. Additionally, the `fprintf` and `fscanf` functions are rather slow when used in a loop. The following code overcomes both issues. This code does not need to be run as part of the tutorial, is also given on MyCourses for future exercises.

```
%clear existing variables, close any figure windows, clear the command line
close all
clear all
clc

% Read image
filename = 'Lena.pgm';
im_orig = imread(filename);
im_orig_info = imfinfo(filename);
numRows = im_orig_info.Height;
numCols = im_orig_info.Width;
imgSize = numRows * numCols;

% Clear any existing serial ports
```

```
delete(instrfind);

% Create a serial port object
serialPort = serial('/dev/ttyUSB1', 'BaudRate', 115200, 'DataBits', 8, 'Parity', 'none',
'StopBit', 1, 'OutputBufferSize', imgSize, 'InputBufferSize', imgSize);

% open the serial port for reading/writing
fopen(serialPort);

% send image to the FPGA

% FULL SPEED

fprintf('Writing Image Data to FPGA... ');
fwrite(serialPort, im_orig(:));

fprintf('Reading back from FPGA... ');
im_final = fread(serialPort, imgSize, 'uint8');

im_final = reshape(im_final, [numRows numCols]);

% close the serial port
fclose(serialPort);

% Analyze and display results
if isequal(im_orig, im_final)
    fprintf('SUCCESS!\n');
    subplot(1,2,1);
    imshow(mat2gray(im_orig));
    title('Original');
    subplot(1,2,2);
    imshow(mat2gray(im_final));
    title('Received');
else
    fprintf('FAILURE!\n');
end
```

Beyond the speed improvements, the code also checks to ensure that the image was not modified during transmission, speeding up the testing process.

6. Save the file with an appropriate name for the part of the exercise.

7. Press F5 to execute the script.

MATLAB Serial Communication Reference Documentation:

`http://www.mathworks.com/help/matlab/serial-port-devices.html`